

Date: _____

PF Theory
Assignment I.

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Part A.

Q1) INPUT N
 $i \leftarrow n$
 WHILE $i \geq 1$
 PRINT i
 $i = i - 1$
 ENDWHILE.

→ N is taken as input. i is used as a counter, decremented till it reaches 1.

Q2) INPUT YEAR
 IF $\text{MOD}(\text{YEAR}, 4) \neq 0$ THEN
 IF $\text{MOD}(\text{YEAR}, 100) \neq 0$ THEN
 IF $\text{MOD}(\text{YEAR}, 400) = 0$ THEN
 PRINT 'Leap Year'
 ENDIF
 ENDIF
 ELSE
 PRINT 'Not a Leap Year'

Q3) INPUT Num

Count $\leftarrow$ 0.

WHILE Num $\neq$ 0 DO

Num $\leftarrow$ Num DIV 10.

Count $\leftarrow$ Count + 1.

ENDWHILE.

PRINT Count

↳ Num is taken as input. Integer Division is performed to remove the last digit. This is done till the zero. Each removed digit is counted

Q4) $i = 1$

WHILE $i \leq 10$ DO.

IF MOD($i, 3$) $\neq$ 0.

$j = 1$.

WHILE $j \leq 10$ DO

PRINT($i * j$)

$j = j + 1$.

ENDWHILE.

ENDIF

$i = i + 1$

Page # ENDWHILE

↳ Two loops are used. Only those no. not divisible by 3 are sent to the inner loop, the inner loop restarts from 1 every time.

Q5) INPUT Num

WHILE Num ≥ 10 .

Sum = 0.

WHILE Num $\neq 0$.

Sum = Sum + MOD (Num, 10).

Num = DIV (Num, 10).

ENDWHILE.

Num = Sum

PRINT Sum.

ENDWHILE